

APPLICATIONS

MODELLING INFLOWS INTO THE GARIEP DAM USING BAYESIAN DYNAMIC MODELS

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Summary: Two dynamic Bayesian models are fitted to the monthly total inflows of the Gariep Dam, making use of the seasonality of the data. Reduced Fourier form models will be used. The first model is a dynamic regression model with AR(1) errors, and the second a state space Kalman filter model. Both models are adequate and with only four parameters good results were found.

1. Introduction

The Gariep Dam was completed and opened in 1971 and is the largest water storage reservoir in South Africa, with a storage volume of approximately 5.5 million m³. It is located in the Orange River, the longest river in South Africa. The dam has the largest hydropower station in South Africa, with a capacity of 360 MW generated by four 90 MW generators. The power station is operated by Eskom, the largest producer of electricity in South Africa. Hydropower is the leading method of generating electricity through renewable energy sources. The modelling of the inflows is of importance, especially in the management of outflows and the planning and forecasting of electricity supplies. Crucially, overflow of the dam and consequent loss of the excess water for electricity generation must be avoided.

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$T = 444$ monthly total observed inflows from 1971 to 2007 are available. The data is highly seasonal with big fluctuations between the high rainfall periods in summer and the low rainfall in winter. Models exploiting the seasonality will be investigated. The models will be fitted to the transformed data, $y = \log(\text{inflow}+1)$. The one is added to prevent the possible taking of the log of zero.

The reduced Fourier form model is discussed in the book of West and Harrison (1997). There are variations on the Fourier form representation of a time series, but all computations to find the most significant harmonics are exactly based on the theory given in their book. A series can be expressed in terms of harmonics and the percentage seasonal variation which is accounted for by a single harmonic can be calculated. In the reduced form model only the few most significant harmonics are used in the model. After experimentation it was found that using the harmonics as independent variables in dynamic regression, the residuals follow an AR(1) process. This will be the first model, and the second model is the dynamic state space form using the most important harmonics. A plot of the series $\{ty\}$, $t = 1, \dots, T$ is shown in figure 1.

The theory of dynamic Bayesian regression models with autocorrelated errors can be found in the books of Zellner (1996) and of Bauwens, Lubrano and Richard (1999). West and Harrison (1997) give a detailed treatment of state space models in a Bayesian context.

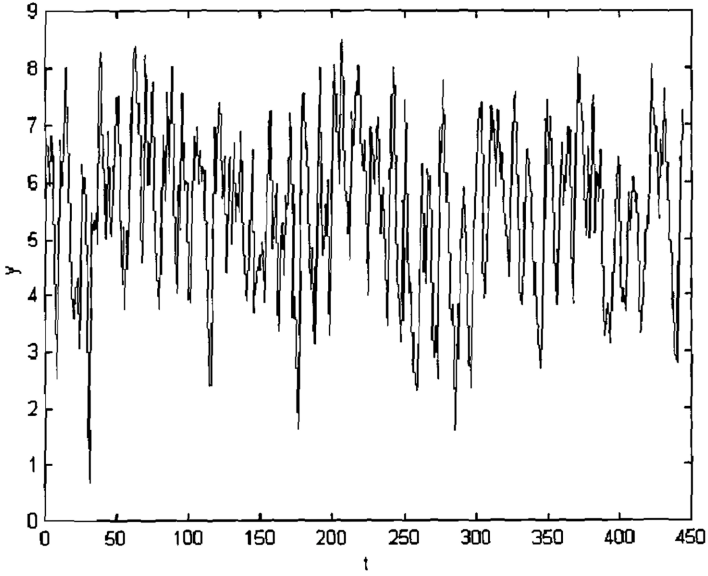


Figure 1. $T = 444$ transformed inflows, 1971–2007.

Let $h = T/2$. Every observation can be expressed as:

$$y_j = a_0 + \sum_{r=1}^h S_{rj},$$

where $\alpha_0 = \bar{y}$, $\alpha = 2\pi/T$. $S_{rj} = a_r \cos(\alpha r j) + b_r \sin(\alpha r j)$ denotes the r th harmonic.

$$a_r = (2/T) \sum_{j=0}^{T-1} y_j \cos(\alpha r j), \quad b_r = (2/T) \sum_{j=0}^{T-1} y_j \sin(\alpha r j),$$

$$a_h = (1/T) \sum_{j=0}^{T-1} (-1)^j y_j, \quad b_h = 0.$$

Let $A_r = (a_r^2 + b_r^2)^{1/2}$ denote the amplitude. For T even the total variation about the mean can be expressed as $(T/2) \sum_{r=1}^h T A_{T/2}^2$. The contribution of each harmonic can be calculated as a ratio of the amplitude to

the total. The first three most important harmonics of the transformed inflow series account for approximately 40% of the variation with periods 1, 12.33 and 2.5 years.

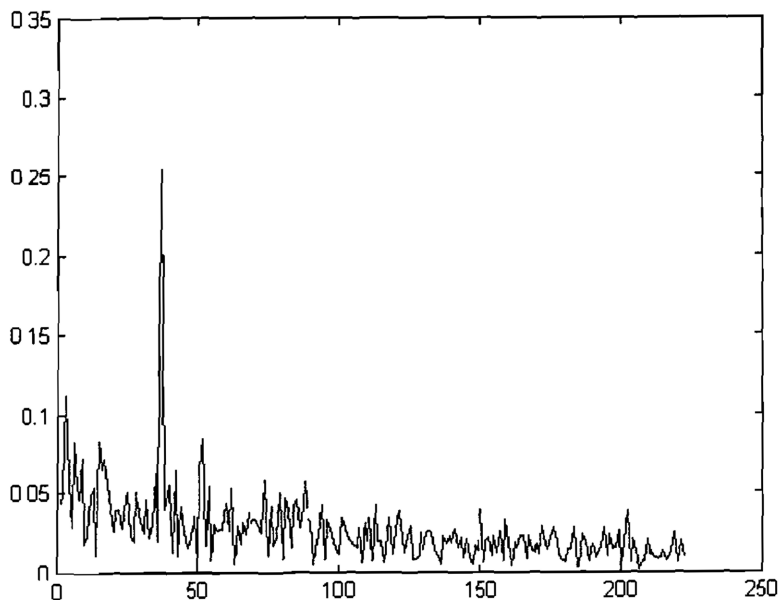


Figure 1b. Proportion of variance accounted for by harmonic components

The period can be found, for example component 37 corresponds to a period of $444/37 = 12$. This is the same form as a periodogram, which is expressed in terms of frequencies. In the following two charts, the data is plotted with the first harmonic component and with the first four harmonic components. The regular seasonal variation presents about 40% of the variation in the series (only the first 4 harmonic components).

The form of the final model is as follows: the time series is the sum of the seasonal variation plus an autoregressive model (AR(1)) to the residuals.

2. Dynamic regression model with AR(1) errors

The notation used as in Zellner (1996) for the theory of dynamic regression models with AR(1) errors will be used. The model is:

$$y_t = \mathbf{x}'_t \boldsymbol{\beta} + u_t, \text{ where } u_t = \rho u_{t-1} + \varepsilon_t, \text{ the } \varepsilon_t \text{'s are i.i.d. } N(0, \sigma^2).$$

Let $\mathbf{y} = X\boldsymbol{\beta} + \mathbf{u}$ and $\mathbf{u} = \rho\mathbf{u}_{-1} + \boldsymbol{\varepsilon}$, where:

$$\mathbf{y}' = (y_1, \dots, y_T) \text{ and } \mathbf{y}'_{-1} = (y_0, \dots, y_{T-1}) \text{ are vectors of } T \text{ observations.}$$

$\mathbf{u}' = (u_1, \dots, u_T)$ and $\mathbf{u}'_{-1} = (u_0, \dots, u_{T-1})$ are vectors of T autocorrelated errors.

$$\boldsymbol{\beta}' = (\beta_1, \dots, \beta_k) \text{ the vector of } k \text{ regression coefficients.}$$

$\boldsymbol{\varepsilon}' = (\varepsilon_1, \dots, \varepsilon_T)$ a vector of T i.i.d. random errors which are assumed to have a normal distribution with mean zero and variance σ^2 , and ρ a scalar parameter.

$$X = \begin{bmatrix} x_{11} \dots x_{1k} \\ \dots \dots \dots \\ x_{T1} \dots x_{Tk} \end{bmatrix} \quad X_{-1} = \begin{bmatrix} x_{01} \dots x_{0k} \\ \dots \dots \dots \\ x_{(T-1)1} \dots x_{(T-1)k} \end{bmatrix}.$$

This model can be written as:

$$\mathbf{y} = \rho\mathbf{y}_{-1} + (\mathbf{X} - \rho\mathbf{X}_{-1})\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$$\mathbf{x}'_t = (1 \cos(\alpha(t-1)r_1) \sin(\alpha(t-1)r_1) \dots \cos(\alpha(t-1)r_m) \sin(\alpha(t-1)r_m))$$

and the model can be written in the form:

$$\mathbf{y}^* = X^*\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \text{ with } \mathbf{y}^* = \mathbf{y} - \rho\mathbf{y}_{-1}, \quad X^* = X - \rho X_{-1}.$$

It will be assumed that there is an intercept and $E(y_0) = \mathbf{x}'_0\boldsymbol{\beta} + M$, M is regarded as a parameter.

The vague prior will be chosen as in Zellner (1996) and is $p(\boldsymbol{\beta}, \rho, \sigma, M) \propto 1/\sigma$. M will be integrated out. Even in the case of the non-informative prior, the resulting posteriors lead to marginal densities where

the normalizing constants must be computed by making use of numerical integration. In the problem under consideration, the sample size is large and it is reasonable to expect that the influence of the prior will be small.

Since there is the complication with the marginal distributions not in known closed forms, several methods were devised to estimate the parameters, especially ρ . One method to estimate ρ is the Cochrane-Orcutt method (Gujarati (1995)), where ρ is estimated repeatedly until it converges. The posterior distributions of the parameters are given in Zellner (1996).

The posterior of β , σ , ρ is:

$$p(\beta, \sigma, \rho | X, y) \propto \frac{1}{\sigma^{T+1}} \exp \left(-\frac{1}{2\sigma^2} [y - \rho y_{-1} - (X - \rho X_{-1})\beta]' [y - \rho y_{-1} - (X - \rho X_{-1})\beta] \right)$$

$$H = X^* X^*, \hat{\beta}(\rho) = H^{-1} X^* y^*.$$

Let $\hat{\epsilon}(\rho) = [y - \rho y_{-1} - (X - \rho X_{-1})\hat{\beta}]'$, then it follows that:

$$p(\sigma | \rho, X, y) \propto \frac{1}{\sigma^{T-k+1} |H|^{1/2}} \exp \left(-\frac{1}{2\sigma^2} (\hat{\epsilon}(\rho))' \hat{\epsilon}(\rho) \right).$$

The conditional posterior of β is multivariate t with $v = T - k$ degrees of freedom:

$$p(\beta | \rho, y) \propto \left\{ v \hat{\sigma}^2 + (\beta - \hat{\beta})' H (\beta - \hat{\beta}) \right\}^{-T/2}.$$

Some remarks can be made about the prior. The proper prior which is normally used in this problem, is the normal inverted gamma prior (NIG) (Bauwens et al., 1999). The unconditional prior of σ^2 is an inverted gamma density with parameters a and b :

$$p(\sigma^2) = \frac{(a/2)^{b/2}}{\Gamma(b/2)} (\sigma^2)^{-(b+2)/2} \exp(-a/2\sigma^2).$$

For this prior the posterior is:

$$p(\beta, \sigma^2, \rho | X, y) \propto \sigma^{-(T+b+3)}$$

$$x \exp \left[-\frac{1}{2\sigma^2} \left(a + (\mathbf{y} - \rho \mathbf{y}_{-1} - (\mathbf{X} - \sigma \mathbf{X}_{-1}) \boldsymbol{\beta})' (\mathbf{y} - \rho \mathbf{y}_{-1} - (\mathbf{X} - \sigma \mathbf{X}_{-1}) \boldsymbol{\beta}) \right) \right].$$

The resulting conditional posterior of $\boldsymbol{\beta}$ given ρ is again a multivariate Student t density with the same expected value as that of the posterior when using a vague prior:

$$p(\boldsymbol{\beta} | \rho, \mathbf{X}, \mathbf{y}) \propto \left\{ \left(a + v\hat{\sigma}^2 \right) + \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}} \right)' H \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}} \right) \right\}^{(T+b+1)/2}.$$

For large T it follows that:

$$T + b + 1 = T \left(1 + \frac{b+1}{T} \right) \sim T,$$

$$\frac{1}{a + v\hat{\sigma}^2} H = \frac{1}{v\hat{\sigma}^2 \left(1 + a/v\hat{\sigma}^2 \right)} H \sim \frac{1}{v\hat{\sigma}^2} H, \quad v = T - k.$$

The vague prior will be used since the sample size $T = 444$ is large and the differences between the point estimates of the two posteriors are negligible.

Suppose now m harmonics $r_1, \dots, r_m$ are used in the reduced form Fourier model. The row of $\mathbf{X}$ corresponding to time t is:

$$\mathbf{x}_t' = (1 \cos(\alpha(t-1)r_1) \sin(\alpha(t-1)r_1) \dots \cos(\alpha(t-1)r_m) \sin(\alpha(t-1)r_m)).$$

For m harmonics the number of parameters including an intercept and excluding ρ is $k = 2m + 1$ and $k + 1$ including ρ . The dynamic regression model was investigated with between 1 and 7 harmonics. In each case the residuals $\{u_t\}$ followed an AR(1) process. For three harmonics, the Durbin-Watson d statistic of the u_t 's is 0.8340, which is highly significant when testing for a positive autocorrelation. The first partial autocorrelation is 0.5746 and the absolute values of the higher order partial autocorrelations are insignificant and less than 0.0949.

Assuming normality of the residuals, the following form of the Schwartz Bayesian Information Criterion (SBC) and the Akaike Information Criterion (AIC) will be used as in Enders (2004):

$$AIC = T \log(\text{sum of squared residuals}) + 2(k + 1)$$

$$SBC = T \log(\text{sum of squared residuals}) + (k + 1)\ln T.$$

According to the SBC criterion one harmonic should be used and the AIC criterion reaches a minimum at more than 7 harmonics. The SBC minimum is 2656.6 and AIC is 2710.5 with 7 harmonics.

The mean squared errors ranged between $\hat{\sigma}^2 = 0.8793$ with 1 harmonic and $\hat{\sigma}^2 = 0.8118$ for 7 harmonics. The estimate of ρ varied between $\hat{\rho} = 0.6212$ for 1 harmonic and $\hat{\rho} = 0.5119$ with 7 harmonics. The residuals of each of these models are white noise with no significant autocorrelations up to lags 10 and tested again up to lags 15 using the Ljung-Box Q statistic (Enders, 2004). The white noise is symmetric, not Gaussian and with a kurtosis of about 3.5.

When one harmonic is used, harmonic 37 corresponding to a period of 12, the results are:

$\hat{\beta}' = (5.4645, 1.321, 0.2779)$, $\hat{\sigma}^2 = 0.8793$, $\hat{\rho} = 0.6212$. In figure 3 the observed and predicted values are shown when three harmonics 37, 3 and 15 are used. The corresponding periods are 12 months, 148 months and 29.6 months. The angular frequencies or 2π times the frequencies, are 0.5236,

0.0425 and 0.2123. The parameter estimates are $\hat{\sigma} = 0.8534$, $\hat{\rho} = 0.5773$ and $\hat{\beta} = (5.4644, 1.1325, 0.2759, -0.3703, 0.1876, -0.0894, 0.3455)$.

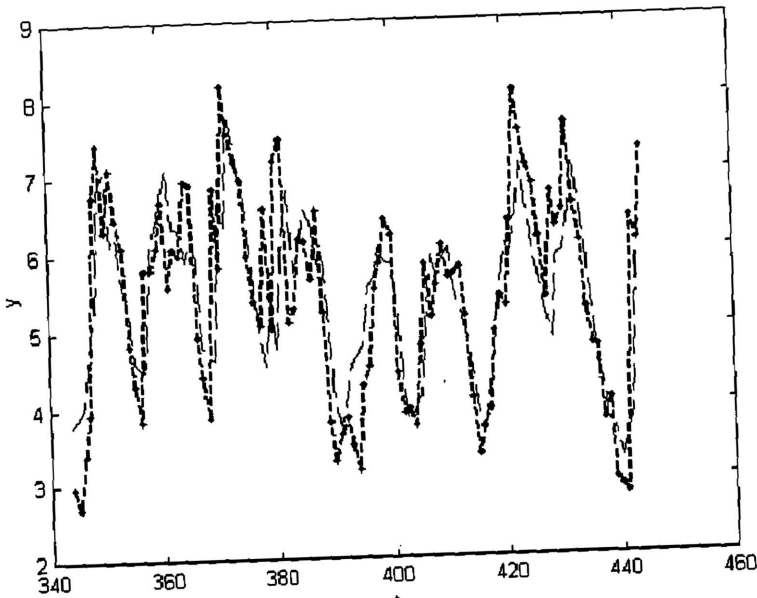


Figure 2. Last 100 observations, observed and predicted. Dashed line denotes the observed values

The cycles are regular, but it can be seen that the largest errors occur in the prediction of the extremes. A histogram of the estimated residuals, $\hat{\varepsilon}$'s, is shown in figure 3. The skewness of the residuals is 0.3720 and kurtosis 3.4869.

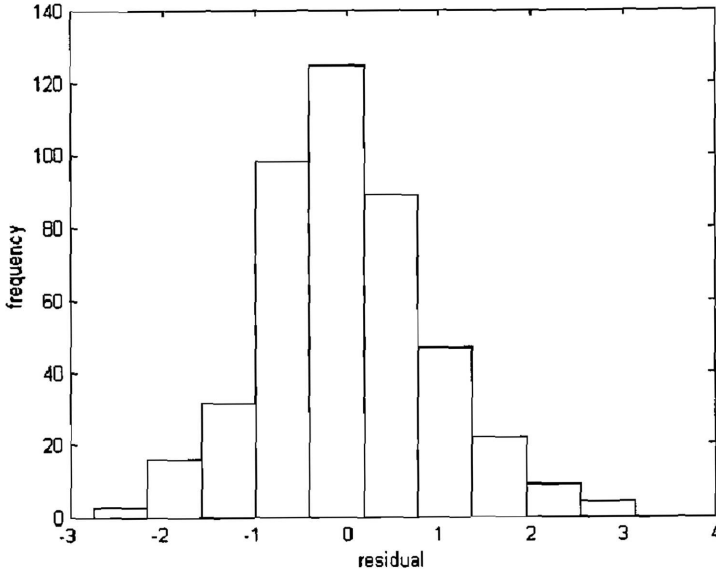


Figure 3. Histogram of estimated residuals

3. Kalman dynamic model

Let D_t denote all the information up to time t . The following dynamic model as discussed in West and Harrison (1997) will be fitted to the series:

Observation Equation: $y_t = x_t' \beta_t + v_t, \quad v_t \sim N(0, V),$

System Equation: $\beta_t = G\beta_{t-1} + w_t, \quad G = I_k, \quad w_t \sim N(0, VW),$

and $\phi = V^{-1}$ with initial information:

$$(\beta_0 | D_0, \phi) \sim N(m_0, VC_0^*), \quad (\phi | D_0) \sim \text{Gamma}\left(\frac{n_0}{2}, \frac{n_0 S_0}{2}\right),$$

the initial prior values m_0, C_0^*, n_0, S_0 are specified. S_0 is a prior estimate of V . These estimates can be found from the analysis in section 2. Under these

conditions the forecast equations are:

$$(Y_t | D_{t-1}) \sim t_{n_{t-1}}(f_t, Q_t),$$

$$(\beta_t | D_{t-1}) \sim t_{n_{t-1}}(\mathbf{a}_t, R_t),$$

where

$$R_t = GC_{t-1} + W_t$$

$$= C_{t-1} + W,$$

$$\mathbf{a}_t = G_t \mathbf{m}_{t-1} = \mathbf{m}_{t-1},$$

$$Q_t = \mathbf{x}_t' R \mathbf{x}_t + S_{t-1}, \quad f_t = \mathbf{x}_t' \mathbf{a}_t.$$

The updating and other equations are given by Harrison and West (1997). The results using only one harmonic are shown in figure 4. The mean square prediction error, calculated from one-step forecasts, is 1.3209. This is larger than the error of the model in section 2, but the main problem is that the series of these errors follows an ARMA type model of at least order 3 for the AR and also the MA part.

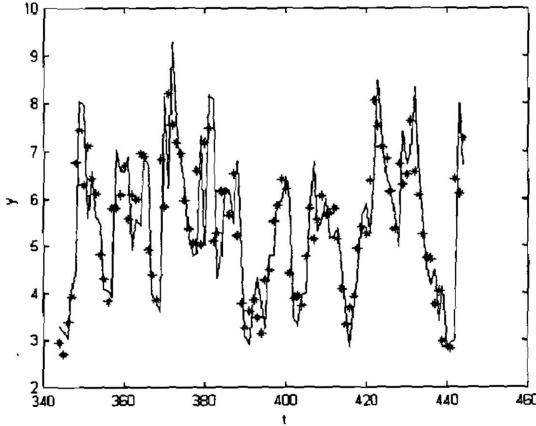


Figure 4. Last hundred One-step predicted values and observed series (*) with 1 harmonic

4. Conclusion

From the viewpoint of parsimony a dynamic regression model with between one and seven harmonics and AR(1) errors would be the best model. When new data becomes available, it is easy to update the parameter estimates and also easy to experiment with other exogenous variables which might be included.

The Box-Cox transformation on the transformed inflows can be investigated. The second highest partial autocorrelation is not significant, but close to being significant. It might be interesting to investigate a model with AR(2) errors, and using the Box-Cox transformation.

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